

S³×S Research — Gates Tracker

Updated: 2026-07-07 · Tests: 2484 passed, 4 skipped · Hard constraints: λ =FREE_COUPLING_PARAMETER · sm_derivation_claimed=False

Gate	Section	Claim (one line)	Status	Test s	Date
G6	Fermion Chain	32-component spinor = 1 SM generation under SO(4)×G	PASS	6/6	2026-06-15
G7	Fermion Chain	KK mass spectrum on S ³ ×S from Lichnerowicz	PASS	8/8	2026-06-15
G8	Fermion Chain	Chirality obstruction on round S ³ ×S	PASS	6/6	2026-06-15
G9	Fermion Chain	Coset chirality: G/SU(3) gives SU(3) action on S	PASS	4/4	2026-06-15
G10	Gauge Structure	S spin connection → SO(6) gauge field	PASS	6/6	2026-06-17
G10b	Gauge Structure	SU(3) embedded in SO(6) explicitly	PASS	5/5	2026-06-18
G11	Gauge Structure	32×32 block generators SU(2)_L, SU(2)_R, SU(3)	PASS	12/12	2026-06-18
G12	Gauge Structure	All 5 SM anomaly cancellations	PASS	5/5	2026-06-18
G13	Gauge Structure	Twisted Dirac index on S ≠ 0	PASS	4/4	2026-06-19
G14	Gauge Structure	Quark color triplet from S spinor basis	PASS	011, 101,	
G15	Quantum Numbers	Hypercharge Y from S geometry	PASS	8/8	2026-06-19
G16	Quantum Numbers	Y = K + (B-L)/2 fully geometric	PASS	8/8	2026-06-19
G17	Quantum Numbers	Q = T + Y; ΣQ=0 for all 32 states	PASS	10/10	2026-06-19
G18	NCG Structure	Spectral triple KO-dim=6; 4 free Yukawa	PASS	8/8	2026-06-19
G19	NCG Structure	(2,2) Higgs bidoublet from D_F	PASS	6/6	2026-06-19
G20	NCG Structure	Yukawa intertwiner dim=4 (8→4 via CPT)	PASS	6/6	2026-06-19
G21	NCG Structure	S necessary: Extended Schur dim check	PASS	5/5	2026-06-19
G22	NCG Structure	NCG first-order condition selects SU(3)×U(1){B-L}	PASS	6/6	2026-06-19
G23	Closing Gates	Chirality from gauge sectors; Witten index=0	PASS	8/8	2026-06-19
G24	Closing Gates	Blind Spectrum: SO(4)×G rep → SM content	PASS	6/6	2026-06-19
G25	Closing Gates	Yukawa Texture: 256→16→4 geometric cascade	PASS	6/6	2026-06-19

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G26	Closing Gates	CCM 2006 comparison: 3 postulates become derived	PASS	5/5	2026-06-19
G28	Spectral Action	Spectral action $\rightarrow$ SM gauge kinetic terms	PASS	8/8	2026-06-19
G29	Spectral Action	Coupling ratio $g_{\text{■}}^2/g_{\text{■}}^2 = 15/(16\pi)$	PASS	6/6	2026-06-20
G27	N_gen NULL	$Z_{\text{■}}$ orbifold on $S_{\text{■}}$ gives $N_{\text{gen}}=3$?	NULL	8/8	2026-06-20
G30	N_gen NULL	$G_{\text{■}}$ -instanton index on $S_{\text{■}}$ gives $\text{ind}=3$?	NULL	34/34	2026-06-20
G31	N_gen NULL	S^3 adjoint bundle $j=1$ gives $\text{ind}=3$?	NULL	12/12	2026-06-20
G33	N_gen NULL	Euler class $c_{\text{■}}(T^{\wedge}\{1,0\}S_{\text{■}})=\chi(S_{\text{■}})$ gives $N_{\text{gen}}=3$?	NULL	10/10	2026-06-20
G34-D1	N_gen NULL	Flux $H_{\text{■}}(S_{\text{■}};\text{■})=\text{■}$ selects $c_{\text{■}}=6$?	WEAK	4/4	2026-06-20
G34-B3	N_gen NULL	WZW $SU(2)_k$ level $k=2$ gives 3 primaries $\rightarrow N_{\text{gen}}=3$?	NULL	6/6	2026-06-20
G34-A2	N_gen NULL	Cobordism $\Omega^{\wedge}\text{Spin}_6=0$ selects $N_{\text{gen}}=3$?	NULL	4/4	2026-06-20
G35	N_gen NULL	NCG $M_{\text{■}}(\text{■})$ dimension = generation counter?	NULL	6/6	2026-06-20
G36	N_gen NULL	K-theory Adams ops select $n=3$?	NULL	8/8	2026-06-20
G37	N_gen NULL	String tadpole on $S^3 \times S_{\text{■}}$ selects $N_{\text{gen}}=3$?	NULL	4/4	2026-06-20
G38	N_gen NULL	Spectral action minimum at $c_{\text{■}}=6$?	NULL	8/8	2026-06-20
G39-B1	N_gen NULL	Pati-Salam $SU(4)$ bundle: $\Lambda^2(T^{\wedge}\{0,1\})$ gives $c_{\text{■}}=6$?	NULL	6/6	2026-06-24
G40-B2	N_gen NULL	$G_{\text{■}} \rightarrow SU(3)$ SSB via $\pi_{\text{■}}(S_{\text{■}})$ gives $c_{\text{■}}=6$?	WEAK	4/4	2026-06-24
G41-B3	N_gen NULL	3 D6-branes: rank-3 gauge bundle selects $c_{\text{■}}=6$?	WEAK	3/3	2026-06-24
G42-B4	N_gen NULL	Green-Schwarz anomaly cancellation selects $c_{\text{■}}=6$?	NULL	4/4	2026-06-24
G43+G48	N_gen NULL	Stable HYM bundles (Harland-Nölle): $c_{\text{■}}=6$ on $S_{\text{■}}$?	NULL	8/8	2026-06-24
G44	N_gen NULL	$D_{\text{■}}$ triality on $S^3 \times S_{\text{■}}$ ($G_{\text{■}}$ holonomy) gives $N_{\text{gen}}=3$?	NULL	6/6	2026-06-20
G45	N_gen NULL	$D_{\text{■}}$ triality on $S^3 \times S_{\text{■}}$ ($SO(8)$) gives $N_{\text{gen}}=3$?	WEAK	4/4	2026-06-20
G46	N_gen NULL	Geometric $\text{■} \otimes \text{■} \otimes \text{■}$ realization gives $N_{\text{gen}}=3$?	NULL	6/6	2026-06-20
T1	N_gen THEOREM	No single-bundle mechanism selects $N_{\text{gen}}=3$ on $S^3 \times S_{\text{■}}$	THEOREM	29/29	2026-06-24
G50	χ -Lemma	χ -lemma: $H^2=H_{\text{■}}=0 \rightarrow c_{\text{■}}=c_{\text{■}}=0 \rightarrow$ single algebraic cause	PROMOTE	10/10	2026-06-20

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G53	Casimir	Casimir energy on $S^3 \times S^1$; $A = \sqrt{\pi}/2$	PARTIAL	12/12	2026-06-21
G54A	Stabilization	Freund-Rubin flux: $V_{\text{flux}} = g^2/g^2$	PASS	8/8	2026-06-21
G57	Stabilization	UV-selection: Casimir half-integer pole $c_{\{1/2\}}=0$ at ρ^*	PASS	6/6	2026-06-21
G60	Stabilization	Minkowski uplift: can NP term alone reach $V=0$?	NULL	6/6	2026-06-21
G62	Stabilization	Zero-fit observables from potential chain	PROMOTE	19/19	2026-06-21
G63	Stabilization	Casimir correction to ρ_{min}	PROMOTE	13/13	2026-06-21
G64	Stabilization	$\rho_{\text{min}}=1.179$ independent of C (g/g ratio)	PROMOTE	23/23	2026-06-21
G65	Stabilization	$\kappa=\rho_{\text{min}}/p^*$ is C-invariant	PROMOTE	24/24	2026-06-21
G66	Stabilization	Analytic $\kappa^2 = (n+1)/n = 7/6$	PROMOTE	25/25	2026-06-21
G67	Three Channels	SO(8) triality $Z \rightarrow 3$ independent channels	PROMOTE	Z-orbit	=3; G68: L≠R (2/3 closed)
G69	Three Channels	CSDR $G/SU(3)$: Spin(6)→SU(3) branching = SM content	PROMOTE	26/26	2026-06-21
G73	Three Channels	$\text{ind}(D_{\{S\}} \otimes S) = 1/\text{channel} \times 3 \text{ channels} = N_{\text{gen}}=3$	PROMOTE	29/29	2026-06-21
G74A	Three Channels	Lichnerowicz+G-Schur: $\dim \ker=1$ EXACTLY per channel	PROMOTE	F	/(R/4)=8/451; safety=5.625x; G-singlet mult=1
G74B	Three Channels	$\text{sign}(\text{ind})=+1 \rightarrow \text{LEFT_HANDED_EXCESS}$	PROMOTE	31/31	2026-06-21
G76	Parameter Audit	Parameter registry: classify all constants by provenance	PASS	—	2026-06-22
G82	Parameter Audit	Canonical mass ratio $m_{\text{mod}}/m_{\text{KK}}$ in Einstein frame	CONDITIONAL	5/5	2026-06-22
AV-G0	Contour AV-2	Source trace: C-H eqs 3.27-3.41 VERIFIED_FROM_PDF	PASS	verified	2026-06-10
AV-G1	Contour AV-2	Two-component first-order system	PASS	24/24	2026-06-10
AV-G2	Contour AV-2	Boundary exponent $g_{l0}=0$; $\text{mixed}_{l0}=\cos\alpha$	PASS	45/45	2026-06-10
AV-E1	Contour AV-2	$\phi \cdot g = \cos\alpha \cdot \sin\alpha = \sin(2\alpha)/2$ EXACT	STRONG_PASS	26/26	2026-06-10
AV-E2	Contour AV-2	Angular CG singlet $C^2=0.5$	PASS	21/21	2026-06-10

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BG-G0	Contour BG-H1	Product Dirac on $S^3 \times S^1$: source trace	PASS	verified	2026-06-10
BG-G1	Contour BG-H1	$D^2 = -(k^2 + p^2) \cdot I$ at machine precision	PASS	58/58	2026-06-10
BG-E1	Contour BG-H1	Discrete KK eigenvalue $k \rightarrow 3/2$ ($N \rightarrow \infty$)	PASS	72/72	2026-06-10
BG-E2	Contour BG-H1	$S^3 \times S^1$ bridge: disorder $W=0.5$ stress test	PASS	67/67	2026-06-15
LB-G0	Contour Lambda-B5	Invariant 1-forms $\notin \text{span}(E_i, E'_i)$	STRUCTURAL_SPLIT	12/12	2026-06-11
LB-G2	Contour Lambda-B5	$\cot(2\alpha)$ is Hopf-frame artifact, not physical	PASS	14/14	2026-06-11
LB-G4	Contour Lambda-B5	λ non-identifiable from S^3 geometry alone	PASS	18/18	2026-06-11
LB-VR	Contour Lambda-B5	$R=\sqrt{2}$ EXACT λ -free ratio from S^3 CG coefficients	PASS	7/7	2026-06-11
S6H-G0	Contour S6-HARM	$SO(6)$ Clifford: $\{\Gamma^a, \Gamma^b\} = 2\delta^{ab}$; 8 weight vectors	PASS	4/4	2026-06-15
S6H-G5	Contour S6-HARM	$\cot \beta_k$ universality: same coeff at 4 levels	PASS	17/17	2026-06-15
G85A	Lambda-Map	Poisson resummation: $\exp(-\lambda/p^2)$ from spectral?	NULL	10/10	2026-06-22
G86A	Lambda-Map	Any geometric T with power-law gives $\exp(-\lambda/p^2)$?	NULL	25/25	2026-06-22
G86B	Lambda-Map	Warp factor $\Omega(y)$ on S^3 gives $\exp(-\lambda/p^2)$?	NULL	8/8	2026-06-22
G90	Lambda-Map	Which NP form $V \propto \exp(-a_p/p^4)$ supports minimum?	PASS	12/12	2026-06-22
G91	Lambda Origin	H1 ($\lambda=1/a$) vs H2 ($\lambda=a/(a+N)$): which gives $\lambda=1/3$?	NULL	8/8	2026-07-05
G100	Audit	G73 "x3 channels" dependency map: what does it rely on?	PASS	6/6	2026-07-01
G102	Triality	Can Spin(8) fiber give third triality channel derivably?	NULL	14/14	2026-07-05
G103	Lambda Origin	KK tower lambda-blindness + UV mechanism proposals	PASS	12/12	2026-07-05
G104	Lambda Origin	H1 vs H2 forward test on $S^2 \times S^1$ (different geometry)	NULL	8/8	2026-07-05
G105	Modulus Mass	Analytic derivation of $m_{\text{mod}} \propto p^{\alpha}$ power law	PROMOTE	18/18	2026-07-06
G106	Modulus Mass	Pre-registered prediction at genuinely new $N=7$ point	PROMOTE	6/6	2026-07-06